

ECON 712: Applied Econometrics

Lecture 16: Differences-in-Differences

Introduction – 2x2 DiD Setup

- Suppose that we have a policy that is implemented at a point in time that affects only a group of individuals
- Also, assume that we have panel data on individuals observed before and after the policy is implemented as well as on individuals not affected by the treatment over the same period
- The goal is to estimate the average **causal effect** of the policy on the treated group

- Let Y_{ist} denote the **outcome variable** for individual i , group s , and period t , where:
 - ▶ $s = 1$: individual i belongs to the **never-treated** group
 - ▶ $s = 2$: individual i belongs to the **eventually-treated** group
 - ▶ $t = 1$: **pre-treatment** period
 - ▶ $t = 2$: **post-treatment** period
- We use the **potential outcomes** notation $Y_{ist}(d)$, where d denotes the treatment status:
 - ▶ $Y_{ist}(0)$: outcome when individual i is **not treated**
 - ▶ $Y_{ist}(1)$: outcome when individual i is **treated**
- The **observed** outcome is $Y_{ist} = Y_{ist}(0) + [Y_{ist}(1) - Y_{ist}(0)] \cdot \mathbf{1}[s = 2, t = 2]$

Average Treatment Effect on the Treated

- The object of interest is the **ATT** — the average causal effect of the policy on the treated group in the post-treatment period:

$$ATT = E[Y_{ist}(1) - Y_{ist}(0) \mid s = 2, t = 2]$$

- Since treatment is received by group $s = 2$ at $t = 2$, the first term is **observed**:

$$E[Y_{ist}(1) \mid s = 2, t = 2] = E[Y_{ist} \mid s = 2, t = 2]$$

- The second term is the **fundamental problem of causal inference** — the counterfactual outcome is never observed:

$$E[Y_{ist}(0) \mid s = 2, t = 2] \quad \longleftarrow \text{unobservable}$$

- We need an assumption to replace this counterfactual with observable quantities

Identification I: Cross-Sectional Comparison

- **Assumption (Random Assignment):** Treatment is independent of potential outcomes, so the never-treated group $s = 1$ provides the counterfactual for $s = 2$:

$$E[Y_{ist}(0) \mid s = 2, t = 2] = E[Y_{ist}(0) \mid s = 1, t = 2] = E[Y_{ist} \mid s = 1, t = 2]$$

- Under random assignment, the ATT is identified by a **cross-sectional comparison** of group means at $t = 2$:

$$ATT = E[Y_{ist} \mid s = 2, t = 2] - E[Y_{ist} \mid s = 1, t = 2]$$

- **Limitation:** Random assignment to treatment is rarely available in observational settings; groups $s = 1$ and $s = 2$ typically differ in ways that also affect $Y_{ist}(0)$

Identification II: Before-After Comparison

- **Assumption (No Time Trend):** The untreated potential outcome of the treated group does not change over time:

$$E[Y_{ist}(0) \mid s = 2, t = 2] = E[Y_{ist}(0) \mid s = 2, t = 1] = E[Y_{ist} \mid s = 2, t = 1]$$

- Under this assumption, the ATT is identified by a **before-after comparison** within the treated group:

$$ATT = E[Y_{ist} \mid s = 2, t = 2] - E[Y_{ist} \mid s = 2, t = 1]$$

- **Limitation:** Many outcomes evolve over time for reasons unrelated to the policy (macro shocks, trends); confounding time variation biases this estimator

Identification III: Parallel Trends

- **Parallel Trends Assumption:** In the absence of treatment, both groups would have experienced the *same* change in outcomes over time:

$$\begin{aligned} E[Y_{ist}(0) \mid s = 2, t = 2] - E[Y_{ist}(0) \mid s = 2, t = 1] \\ = E[Y_{ist}(0) \mid s = 1, t = 2] - E[Y_{ist}(0) \mid s = 1, t = 1] \end{aligned}$$

- Since $s = 1$ is never treated and $s = 2$ is untreated at $t = 1$, the right-hand side is **fully observable**. Solving for the counterfactual:

$$\begin{aligned} E[Y_{ist}(0) \mid s = 2, t = 2] = E[Y_{ist} \mid s = 2, t = 1] \\ + \underbrace{(E[Y_{ist} \mid s = 1, t = 2] - E[Y_{ist} \mid s = 1, t = 1])}_{\text{common time trend}} \end{aligned}$$

- This pins down the **unobservable** counterfactual $E[Y_{ist}(0) \mid s = 2, t = 2]$ using only observed data

Identification III: The DiD Estimator

- Substitute the counterfactual into the ATT:

$$\begin{aligned} \text{ATT} &= E[Y_{ist} \mid s = 2, t = 2] - E[Y_{ist}(0) \mid s = 2, t = 2] \\ &= E[Y_{ist} \mid s = 2, t = 2] - E[Y_{ist} \mid s = 2, t = 1] \\ &\quad - \left(E[Y_{ist} \mid s = 1, t = 2] - E[Y_{ist} \mid s = 1, t = 1] \right) \end{aligned}$$

- Every term on the right is **observable**, so the ATT is identified:

$$\begin{aligned} \text{ATT} &= \underbrace{\left(E[Y_{ist} \mid s = 2, t = 2] - E[Y_{ist} \mid s = 2, t = 1] \right)}_{\Delta \text{ treated}} \\ &\quad - \underbrace{\left(E[Y_{ist} \mid s = 1, t = 2] - E[Y_{ist} \mid s = 1, t = 1] \right)}_{\Delta \text{ control}} \end{aligned}$$

- The DiD estimator **differences out** the common time trend, leaving only the causal effect of the policy on the treated group

Example: Card and Krueger (1994)

- The correctly specified OLS regression is an interaction with time and group fixed effects:

$$Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ \times d)_{st} + \varepsilon_{ist}$$

- NJ_s : dummy equal to 1 if observation is from New Jersey
 - d_t : dummy equal to 1 if observation is from November (post period)
- This equation takes the following values:

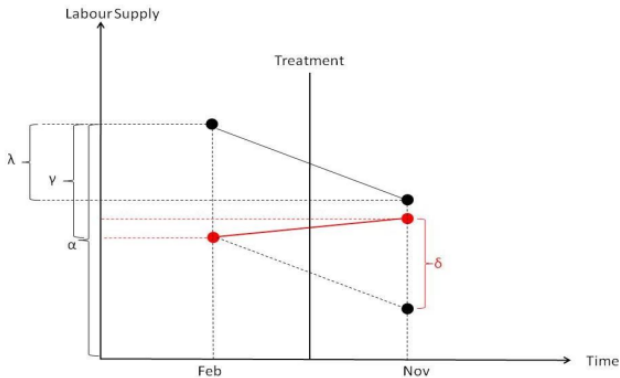
	Pre (Feb)	Post (Nov)
PA (control)	α	$\alpha + \lambda$
NJ (treated)	$\alpha + \gamma$	$\alpha + \gamma + \lambda + \delta$

- The DiD estimator recovers δ :

$$\hat{\delta} = (\bar{Y}_{NJ,Post} - \bar{Y}_{NJ,Pre}) - (\bar{Y}_{PA,Post} - \bar{Y}_{PA,Pre})$$

Example: Card and Krueger (1994)

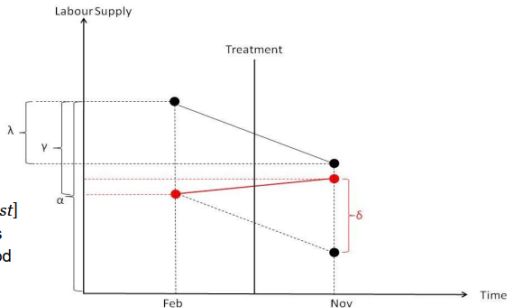
$$Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ \times d)_{st} + \varepsilon_{ist}$$



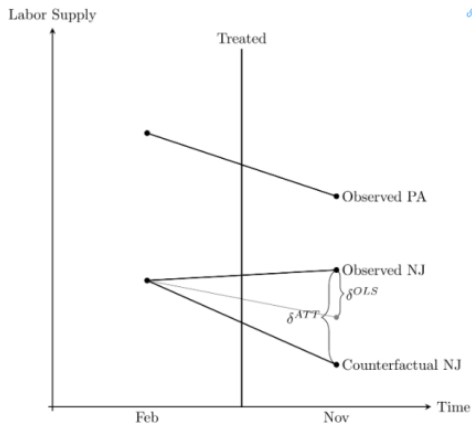
Example: Card and Krueger (1994)

$$Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ \times d)_{st} + \varepsilon_{ist}$$

Notice how OLS is “imputing” $E[Y^0|D = 1, Post]$ for the treatment group in the post period? It is only “correct”, though, if parallel trends is a good approximation



Card and Krueger (1994)– what if parallel trends fail?



Conclusion

- DiD **relaxes** the two strong assumptions seen earlier:
 - ▶ No need for **random assignment** across groups
 - ▶ No need for **no time trends** within the treated group
- Instead, it imposes only **parallel trends**: treated and control groups would have followed the same trend absent treatment — a weaker condition
 - ▶ Parallel trends can also fail if treatment is correlated with trends
 - ▶ In such cases a **triple differences**, which extends the DiD along the same lines, might be able solve endogeneity issues
- A critical practical issue arises when **treatment is assigned at the group level** (e.g. state, industry) but the regression is run at the individual level
 - ▶ In that case, all individuals within a group share the same treatment status, inducing **within-group correlation** in the residuals, so **cluster standard errors should be clustered at the level of treatment assignment**
 - ▶ **Card & Krueger (1994)** is an example of this problem, with only two states, clustering is infeasible, which is a fundamental limitation of the design